

parking space management system is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

Use IC base for the microcontroller. Before inserting the IC, check the supply voltage. Orient the infrared LEDs and photo-transistors such that these directly face each other. House the photo-transistors in suitable enclosures to protect these from ambient light. Set contrast-control VR1 for proper text display on the LCD.

Functioning

One IR transmitter-receiver pair (TX1 and RX1) is mounted on walls before the entry barrier gate (say, IN) and the

other transmitter-receiver pair (TX2 and RX2) on walls before the exit barrier (as shown in Fig. 1). The transmitter and receiver should be mounted about two metres apart.

As the parking lot has the capacity of a hundred cars, initially, the LCD shows '100', indicating that the parking lot has space available for a hundred cars. When a car comes to interrupt the IR signal between transmitter TX1 and receiver RX1, the barrier opens and car is allowed inside the parking lot. The capacity count now decrements by one and the LCD shows '99' along with 'welcome' message.

When a car leaves the parking lot to intercept the infrared signal between transmitter TX2 and receiver RX2, the exit gate opens and the LCD shows

'Bye-bye' message. The microcontroller updates the available space by incrementing the capacity counter by '1.' The number of parking spaces available is displayed on the LCD.

Software

The software for the microcontroller-based parking system is written in 'C' language and compiled using Keil µVision4 compiler. The generated hex code is burnt into the microcontroller using a suitable programmer.

EFY note. The source code of this article is available on www.efymag.com website.

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PARKSPACE.C

```
#include<reg51.h>
#define LCD P0
sbit rs = P2^5;
sbit rw = P2^6;
sbit en = P2^7;
sbit IN_COMING = P1^0;
sbit OUT_GOING = P1^1;
sbit POL = P3^7;
char z;
int TOTAL_CAR;
void MSdelay(unsigned int itime)
{
    unsigned int i,j;
    for(i=0;i<itime;i++)
        for(j=0;j<125;j++);
}
void lcdcmd(unsigned char value)
{
    LCD = value;
    rs = 0;
    rw = 0;
    en = 1;
    MSdelay(1);
    en = 0;
    return;
}
void lcdwrt(unsigned char value)
{
    LCD = value;
    rs = 1;
    rw = 0;
    en = 1;
    MSdelay(1);
    en = 0;
    return;
}
void LCD_INIT()
{
    lcdcmd(0x38);
    MSdelay(1);
    lcdcmd(0x01);
    MSdelay(1);
    lcdcmd(0x00);
    MSdelay(1);
    lcdcmd(0x14);
    MSdelay(1);
    lcdcmd(0x80);
    MSdelay(1);
}
void ADC(unsigned int value)
{
    unsigned char x,d1,d2,d3;
    x = value / 10;
    d1 = value % 10;
    d2 = x % 10;
    d3 = x / 10;
    lcdcmd(0x02);
    lcdwrt(d3 | 0x30);
    lcdwrt(d2 | 0x30);
    lcdwrt(d1 | 0x30);
    return;
}
void main(void)
{
    const char code str1[] = "WELCOME TO CAR";
    const char code str2[] = "PARKING WATCHER";
    const char code str3[] = "SCANNING WHO'S COMING AND GOING";
    const char code str4[] = "No. LEFT:";
    const char code str5[] = "Welcome ----->";
    const char code str6[] = "Bye Bye <-----";
    const char code str7[] = "-----FULL-----";
    P0 = 0xFF;
    P1 = 0xFF;
    P2 = 0xFF;
    P3 = 0xFF;
    IE = 0x00;
    LCD_INIT();
    TOTAL_CAR=100; //PARKING ALLOWED UPTO 100 CARS
    POL=0;
    for(z=0; z<17; z++)
    {
        LCD = str1[z];
        lcdwrt(LCD);
    }
    lcdcmd(0x0C);
    MSdelay(1);
    for(z=0; z<15; z++)
    {
        LCD = str2[z];
        lcdwrt(LCD);
    }
    MSdelay(200);
    lcdcmd(0x01);
    while(1)
    {
        if(IN_COMING==0 & OUT_GOING==0)
        {
            lcdcmd(0x80);
            MSdelay(1);
            for(z=0; z<31; z++)
            {
                LCD = str3[z];
                lcdwrt(LCD);
            }
            lcdcmd(0x1C);
            MSdelay(50);
            if(IN_COMING==1)
            {
                //IF CAR IS COMING
                lcdcmd(0x01);
                lcdcmd(0x80);
                MSdelay(1);
                for(z=0; z<17; z++)
                {
                    LCD = str4[z];
                    lcdwrt(LCD);
                }
                lcdcmd(0x0C);
                MSdelay(1);
            }
            if(TOTAL_CAR==0)
            {
                LCD = str5[z];
                lcdwrt(LCD);
            }
            if(TOTAL_CAR==100)
            {
                LCD = str6[z];
                lcdwrt(LCD);
            }
            if(TOTAL_CAR==1)
            {
                LCD = str7[z];
                lcdwrt(LCD);
            }
        }
        if(OUT_GOING==1)
        {
            TOTAL_CAR = TOTAL_CAR-1;
            if(TOTAL_CAR<0) TOTAL_CAR=0;
            ADC(TOTAL_CAR);
            while(OUT_GOING==1);
        }
        if(IN_COMING==1)
        {
            TOTAL_CAR = TOTAL_CAR+1;
            if(TOTAL_CAR>100) TOTAL_CAR=100;
            ADC(TOTAL_CAR);
            while(IN_COMING==1);
        }
        POL=0;
    }
}
```